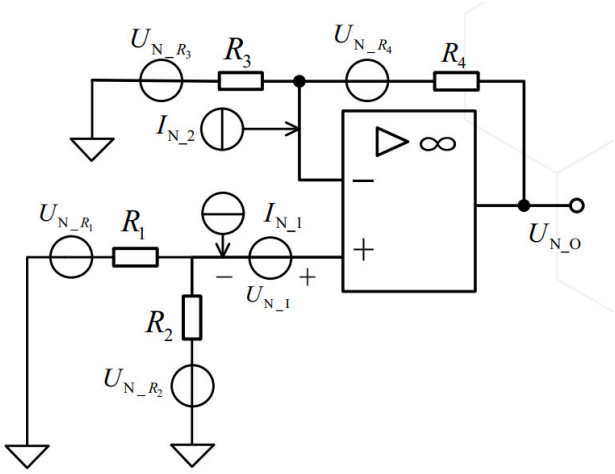
第七次作业

1、电路如图所示，运放为 AD8675，电阻分别为 R1=1kΩ，R2=5kΩ，R3=1kΩ，R4=5kΩ，求电路的输出噪声电压有效值。



解答：通过查询 AD8675 数据手册可知

$GBP = 10MHz$， $en = 2.8nV/\sqrt{Hz}$， $in = 0.3pA/\sqrt{Hz}$， $k = 1.38*10^{-23}J/K$

$Dv-1Hz = 6.5nV/\sqrt{Hz}$， $T = 25^{\circ}C$， $f_c = GBP*\beta = 1.67MHz$

$f_B = 1.57f_c = 2.62MHz$， $f_A = 0.262MHz$

$K = en = 2.8nV/\sqrt{Hz}$， $C = \sqrt{6.5^2 - 2.8^2}nV/\sqrt{Hz} = 5.87nV/\sqrt{Hz}$

$$U_{N-wh} = k*\sqrt{f_B - f_A} \approx 4299.62nV$$

$$U_{N-1/f} = C*\sqrt{1Hz}*\sqrt{\ln\frac{f_B}{f_A}} = 8.91nV$$

$$U_{N-1} = \sqrt{U_{N-wh}^2 + U_{N-1/f}^2} = 4299.63nV$$

四个电阻的热噪声为：

$$U_{N-R_1} = \sqrt{4kTR_1f_B} = 6565nV$$

$$U_{N-R_2} = \sqrt{4kTR_2f_B} = 14680nV$$

$$U_{N-R_3} = \sqrt{4kTR_3f_B} = 6565nV$$

$$U_{N-R_4} = \sqrt{4kTR_4f_B} = 14680nV$$

在本例中，电流噪声极小，为 pA 量级，远远小于电压噪声，故可忽略不计；其余输出噪声分别为：

$$u_1 = \left(1+\frac{R_4}{R_3}\right)U_{N-1} = 25797.78nV$$

$$u_2 = \left(1+\frac{R_4}{R_3}\right)\frac{R_2}{R_1+R_2}U_{N-R_1} = 32825nV$$

$$u_3 = \left(1+\frac{R_4}{R_3}\right)\frac{R_1}{R_1+R_2}U_{N-R_2} = 14680nV$$

$$u_4 = -\frac{R_4}{R_3}U_{N-R_3} = -32825nV$$

$$u_5 = U_{N-R_4} = 14680nV$$

输出总噪声为：

$$u_{总} = \sqrt{u_1^2+u_2^2+u_3^2+u_4^2+u_5^2} = 46.6\mu V$$